Heparin (for anticoagulation)

Neonatal

FORM Ampoule containing 5000units in 5ml (preservative free)

INDICATION Treatment and prophylaxis of thrombosis/embolism

DOSE RANGE

TREATMENT SCHEDULE:

AGE	DOSE	FREQUENCY	ROUTE
Birth – 6 months	**IF APTT <50s**: Initial loading dose of 75units/kg **(DO NOT LOAD IF JUST RETURNED FROM THEATRE)** Then immediately after run infusion at 25units/kg/hour and check APTT in 4 hours (see monitoring below) **IF APTT 50-80s:** Initially run infusion at 25units/kg/hour and check APTT in 4 hours (see monitoring below)	Loading Dose over 10 minutes Continuous Infusion Continuous Infusion	IV IV IV

*Bolus doses may be required throughout infusion based on APTT results – see monitoring table below

PROPHYLAXIS SCHEDULE:

AGE	DOSE	FREQUENCY	ROUTE
Birth – 6 months	20units/kg/hour Loading dose not required	Continuous infusion	IV

PRESCRIPTION OF CONTINUOUS INFUSION

1000units/kg of heparin diluted to 50ml with sodium chloride 0.9%

This gives;
20units/kg/hr at 1ml/hr
25units/kg/hr at 1.25ml/hr

RECONSTITUTION

Already in solution.

DILUTION

For Loading Dose and Bolus Doses

Draw up the required volume of Heparin 1000units/ml. Can be administered either undiluted or further dilute with a convenient volume of sodium chloride 0.9%. Administer over ten minutes

For Continuous Infusion

1ml x weight (kg) is the number of ml of heparin 1000units/ml to be diluted up to 50ml total with sodium chloride 0.9% (equivalent to 1000units/kg in 50ml)

METHOD OF ADMINISTRATION

For Loading dose:
Give dose over ten minutes

For continuous Infusion:
By continuous intravenous infusion, rate adjusted according to APTT if necessary as below.

MONITORING

TREATMENT
Aim for an APTT of 60-90 unless specific instructions otherwise. Adjust rate as follows:

APTT (s)	Bolus (units/kg)	Stop infusion	Rate Change	Recheck APTT (hr)
<50	50units/kg		+ 10%	4
50 - 59				4
60 - 89			No change	24 hours
90 - 99			- 10%	4
100 - 120		30min		4
>120		60min	- 15%	4

- Any change in heparin dosage should be followed by a repeat clotting profile after 4 hours.
- The need for more than 40units/kg/hr of heparin to achieve target APTT should be escalated to the Neonatal Consultant and discussion with Cardiology team is recommended.
- Consider measuring anti-Xa level or anti-thrombin III level. This will be age dependent. It is not unusual for a child <1 year old to require 40units/kg/hr in order to achieve therapeutic APTT levels.
- It is mandatory to measure APTT level 4 hours after the loading dose, 4 hours after any dose change then at least daily when in the therapeutic range.
- Once targetAPTT is achieved and the heparin infusion rate is stable, checking clotting profile once daily is acceptable

Platelets: Check daily. If platelets drop by >50% from baseline consider Heparin Induced Thrombocytopenia (HIT), this is more likely after 5-10 days of treatment. However there are many other reasons for thrombocytopenia (NEC /infection etc). Notify consultant.
Consider sending HIT antibody screen.

PROPHYLAXIS
Clotting profile (APTT, PT and fibrinogen) should be checked as follows:

- On arrival back from theatre
- 4 hours after INITIAL commencement of heparin infusion
- 1 hour after a syringe is changed (APTT needed only, upper limit of 80 is acceptable)
- Daily whilst on heparin infusion (accepting limits of <90)

Platelets: As above

COMPATIBILITY

Solution compatibility	Sodium chloride 0.9%, Sodium Chloride 0.45%, Glucose 5%, Glucose 10%
Solution incompatibility	No information
IV Line compatibility	Adrenaline, acyclovir, benzylpenicillin, calcium gluconate, clonidine, dexamethasone, dopamine, flucloxacillin, fluconazole, furosemide, ganciclovir, hydrocortisone sodium succinate insulin, linezolid, magnesium sulphate, meropenem, metronidazole, midazolam, milrinone, morphine, noradrenaline, piperacillin/ tazobactam, potassium chloride, sodium bicarbonate, sodium nitroprusside,

	vecuronium
IV Line incompatibility	Amiodarone, ciprofloxacin, dobutamine, gentamicin, , labetolol, omeprazole, vancomycin

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
See Summary of Product Characteristics and most recent edition of BNF for Children
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FURTHER INFORMATION

The preservative free formulation is the preferred formulation of choice in neonates. Other formulations contain benzyl alcohol.
Benzyl alcohol has been associated with a fatal toxic syndrome in preterm neonates.
These formulations should not be used in neonates unless there is no alternative.

STORAGE Store below 25°C

LICENSED STATUS Licensed in neonates for treatment and prophylaxis of thrombotic episodes

APPLICABLE POLICIES **Applicable GG&C policies:**
Anticoagulation and antiplatelet therapy in paediatric cardiac patients (1180) | NHSGGC

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications. This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement. For further advice please contact your clinical pharmacist or pharmacy department